NOTEBOOK: NON-COMMUTATIVE AND NON-HILBERT STRUCTURES ATTACHED TO THE ORDINAL–MÖBIUS PAIRING

Contents

1. Introduction 1
2. A naive quantum ansatz, and why it fails 2
3. An elementary obstruction to exact canonical conjugacy 2
4. A number-phase realization on $\ell^2(\mathbb{Z})$ 3
4.1. Construction 3
5. A non-Hilbert-space realization 3
5.1. Ambient space 3
5.2. The commutator 4
5.3. The escape from Hilbert space 4
6. A Bost–Connes-type flow on the Fréchet space 5
6.1. The ring homomorphism property 5
6.2. Hamiltonian and time evolution 5
6.3. The commutator with the Hamiltonian 6
6.4. A formal partition function 6
7. Related structures in the literature 7
8. Open questions 7
Notebook History 7
Notebook History 7
References 8

Remark 0.1. This notebook is exploratory in nature. Its purpose is to investigate whether the pair $(n, \mu(n))$ – the ordinal index and the value of the Möbius function at that index – can be organized into a coherent operator-theoretic structure exhibiting a genuine, non-statistical incompatibility analogous to the position-momentum complementarity of quantum mechanics.

Several sections contain formal manipulations, heuristic arguments, and speculative interpretations. Whenever a statement is rigorous it is explicitly identified as such (Definition, Proposition, Lemma, Corollary), with a full derivation. All remaining claims are tentative, heuristic, or conjectural, and are marked as such.

The scope of this notebook is the algebraic and operator-theoretic structure of the ordinal–Möbius pairing itself. It does not address the asymptotic behaviour of the Mertens function $M(x) = \sum_{k \le x} \mu(k)$ or the Riemann Hypothesis; no result below bears on either.

1. Introduction

The Möbius function $\mu : \mathbb{N} \to \{-1, 0, 1\}$ is defined by $\mu(1) = 1$, $\mu(n) = (-1)^{\omega(n)}$ if n is squarefree with $\omega(n)$ distinct prime factors, and $\mu(n) = 0$ otherwise. The present notebook records an attempt to attach to the pair $(n, \mu(n))$ an operator-theoretic structure in which n (the ordinal) plays the role of a *position* variable and $\mu(n)$ (or a quantity built from it) plays the role of a *momentum*-type

variable, in such a way that knowledge of one genuinely precludes exact knowledge of the other – not as a statement about measurement precision, but as a structural fact about the objects involved.

The motivating question, stated informally, is:

Question 1.1. Can the ordinal n and the Möbius value $\mu(n)$ be realized as a canonically conjugate pair, in a sense that reproduces the qualitative content of Heisenberg complementarity (an intrinsic impossibility, not a statistical spread), while remaining fully precise and auditable at each step?

Section 2 records a first, heuristic attempt ("MQMNL") and explains why it does not answer Question 1.1: it postulates an uncertainty relation without deriving it from the algebra of the operators involved. Section 3 isolates a precise obstruction to the most literal reading of the question. Section 4 gives a Hilbert-space construction, adapted from the number-phase formalism of quantum optics, in which a genuine (derived, not postulated) complementarity holds between the ordinal and a phase variable, with μ recovered as a coarse-grained observable. Section 5 gives a second, non-Hilbert-space construction, built directly from the Dirichlet convolution ring of arithmetic functions, in which the incompatibility between n and μ is an algebraic fact with no probabilistic content at all. Section 7 discusses the closest structures found in the literature. Section 8 lists open questions.

2. A naive quantum ansatz, and why it fails

Remark 2.1 (Failed attempt). An initial heuristic proposal ("Möbius Quantum Model, Non-Localizable", MQMNL) represents the partial sum $M(n) = \sum_{k\le n} \mu(k)$ as a state $|\Psi_n\rangle = \sum_{m\in\mathbb{Z}} a_m^{(n)} |m\rangle$ in $\ell^2(\mathbb{Z})$, evolving by

$$|\Psi_n\rangle = T_{\mu(n)}|\Psi_{n-1}\rangle, \qquad T_{+1}|m\rangle = |m+1\rangle,\ T_0 = \mathrm{Id},\ T_{-1}|m\rangle = |m-1\rangle, \tag{2.1}$$

and postulates an uncertainty relation

$$\Delta\mathcal{M}_n \, D_{\theta,n} \ge \kappa \tag{2.2}$$

for some constant $\kappa > 0$, where D_θ is a dispersion of a conjugate phase variable.

This does not answer Question 1.1, for an identifiable reason: the operators T_{+1}, T_0, T_{-1} in (2.1) all commute with one another, being translations on $\mathbb{Z}$, which forms an abelian group. A translation does not broaden a distribution, so the evolution (2.1) alone provides no mechanism that could force (2.2). The relation (2.2) is an additional hypothesis, not a consequence of (2.1). This approach is therefore recorded here as a failed attempt: it does not derive the complementarity it sets out to model.

3. An elementary obstruction to exact canonical conjugacy

The classical Heisenberg relation asks for operators $\hat{A}, \hat{B}$ with $[\hat{A}, \hat{B}] = ic\hat{I}$, $c \neq 0$ a real scalar. It is standard that such a relation cannot be realized by two *bounded* operators on a Hilbert space.

Remark 3.1 (Standard background). If $\hat{A}, \hat{B}$ are bounded operators on a Hilbert space and $[\hat{A}, \hat{B}] = c\hat{I}$ for a nonzero scalar c, no such pair exists; equivalently, at least one of the two operators satisfying an exact canonical commutation relation must be unbounded [1, 2].

For the ordinal–Möbius pair a stronger, elementary, and self-contained statement holds, not requiring both operators to be bounded.

Definition 3.2. Let $S \subseteq \mathbb{Z}$ be a set with no repeated elements (e.g. $S = \mathbb{N}$), and let $\{|n\rangle\}_{n\in S}$ be an orthonormal family. The *position operator* $\hat{n}$ is defined on the algebraic span of this family by $\hat{n}|n\rangle = n|n\rangle$.

Proposition 3.3. *Let $\hat{n}$ be as in Definition 3.2. There is no linear operator $\hat{B}$, defined at least on the span of $\{|n\rangle\}_{n\in S}$ (bounded or not), satisfying $[\hat{n}, \hat{B}] = ic\hat{I}$ for a nonzero scalar c.*

Proof. Write $B_{mn} = \langle m|\hat{B}|n\rangle$. Taking matrix elements of $[\hat{n}, \hat{B}] = ic\hat{I}$ between $\langle m|$ and $|n\rangle$ gives

$$(m - n)\, B_{mn} = ic\, \delta_{mn}. \tag{3.1}$$

For $m \neq n$, since S has no repeated elements $m - n \neq 0$, so (3.1) forces $B_{mn} = 0$: any $\hat{B}$ satisfying the relation is diagonal in this basis. For $m = n$, the left side of (3.1) is 0, while the right side is $ic \neq 0$, a contradiction. □

Corollary 3.4. *Let $\hat{\mu}$ be any operator with $\hat{\mu}|n\rangle = \mu(n)|n\rangle$ absorbed into a basis change, or more generally any operator with spectrum contained in $\{-1, 0, 1\}$. Then $\hat{\mu}$ cannot satisfy $[\hat{n}, \hat{\mu}] = ic\hat{I}$ for any nonzero c. In particular, the literal reading of Question 1.1 – $\mu(n)$ as the exact canonical momentum conjugate to n, in the sense $[\hat{n}, \hat{\mu}] = i\hbar\hat{I}$ – is not realizable, regardless of boundedness.*

Proof. Immediate from Proposition 3.3, which places no boundedness hypothesis on $\hat{B}$. □

Remark 3.5. Corollary 3.4 rules out the most literal form of Question 1.1. The two remaining sections propose two different, weaker, but fully rigorous readings of the question, one Hilbert-space based (Section 4) and one not (Section 5).

4. A number-phase realization on $\ell^2(\mathbb{Z})$

4.1. **Construction.** This construction adapts the number-phase formalism of quantum optics [3, 4] – there developed for the photon-number operator and its conjugate phase – to the ordinal variable n.

Definition 4.1. On $\mathcal{H} = \ell^2(\mathbb{Z})$ with orthonormal basis $\{|n\rangle\}_{n\in\mathbb{Z}}$, define the position operator $\hat{n}$ as in Definition 3.2 and the unitary shift operator $\hat{S}$ by $\hat{S}|n\rangle = |n + 1\rangle$.

Proposition 4.2 (Weyl relation)**.**

$$[\hat{n}, \hat{S}] = \hat{S}. \tag{4.1}$$

Proof. $\hat{n}\hat{S}|n\rangle = \hat{n}|n + 1\rangle = (n + 1)|n + 1\rangle$, and $\hat{S}\hat{n}|n\rangle = n|n + 1\rangle$. Subtracting gives $[\hat{n}, \hat{S}]|n\rangle = |n + 1\rangle = \hat{S}|n\rangle$ for every basis vector, hence (4.1). □

Remark 4.3. Relation (4.1) does not conflict with Proposition 3.3: the right-hand side is $\hat{S}$, not a nonzero scalar multiple of $\hat{I}$.

By Pontryagin duality between $\mathbb{Z}$ and the circle $\mathbb{T} = [-\pi, \pi)$, write $|\theta\rangle = \sum_{n\in\mathbb{Z}} e^{in\theta}|n\rangle$ (formally); then $\hat{S} = e^{i\hat{\theta}}$ in this representation, and $\hat{\theta}$ is the genuine conjugate variable to $\hat{n}$.

Definition 4.4. Fix a measurable partition $\mathbb{T} = \Delta_{-1} \sqcup \Delta_0 \sqcup \Delta_{+1}$ into three arcs. Define the bounded self-adjoint operator

$$\hat{\mu} \;=\; (-1)\,\Pi_{\Delta_{-1}} + 0 \cdot \Pi_{\Delta_0} + (+1)\,\Pi_{\Delta_{+1}}, \tag{4.2}$$

where Π_{Δ_c} is the spectral projection of $\hat{\theta}$ onto the arc Δ_c.

The partition in Definition 4.4 is a free choice of the construction; nothing below determines it from the arithmetic of μ.

Proposition 4.5. *For every $n \in \mathbb{Z}$ and every $c \in \{-1, 0, 1\}$,*

$$P(\hat{\mu} = c \mid \hat{n} = n) = \frac{|\Delta_c|}{2\pi}, \tag{4.3}$$

independently of n.

Proof. In the θ-representation, $|n\rangle$ corresponds to the function $\theta \mapsto e^{-in\theta}$, of constant modulus 1 on $\mathbb{T}$. Hence $P(\hat{\mu} = c \mid \hat{n} = n) = \int_{\Delta_c} |e^{-in\theta}|^2 \frac{d\theta}{2\pi} = |\Delta_c|/2\pi$, which does not depend on n. □

Corollary 4.6. *If every Δ_c has $0 < |\Delta_c| < 2\pi$, then $\mathrm{Var}(\hat{\mu} \mid \hat{n} = n) > 0$ for every n: there is no exceptional value of n for which the outcome of $\hat{\mu}$ becomes determined.*

Proof. Immediate from (4.3): the three probabilities are positive constants independent of n, so the variance is the same positive constant for every n. □

Remark 4.7. In ordinary quantum mechanics, non-commuting observables can share an exceptional common eigenstate when the commutator's expectation value vanishes there (e.g. the zero-angular-momentum state for $\hat{L}_x, \hat{L}_y, \hat{L}_z$). Corollary 4.6 shows that this pair, $(\hat{n}, \hat{\mu})$, has no analogous exceptional state: the incompatibility holds uniformly.

5. A non-Hilbert-space realization

5.1. **Ambient space.**

Definition 5.1. Let $A = \mathbb{C}^{\mathbb{N}}$, the space of all sequences $a : \mathbb{N} \to \mathbb{C}$, equipped with the topology of pointwise convergence. A is a Fréchet space; it carries no inner product and is not normable in general.

Definition 5.2. For an arithmetic function $f : \mathbb{N} \to \mathbb{C}$, the Dirichlet convolution operator $T_f : A \to A$ is

$$(T_f a)(n) = \sum_{d|n} f(d)\, a(n/d). \tag{5.1}$$

The set of arithmetic functions with pointwise addition and Dirichlet convolution forms a commutative ring, isomorphic to a formal power series ring over $\mathbb{C}$ in countably many indeterminates and, in particular, a unique factorization domain [5].

Definition 5.3. The position operator $\hat{N} : A \to A$ is $(\hat{N}a)(n) = n \cdot a(n)$.

Note that T_ζ (convolution with the constant function $\zeta \equiv 1$) and T_μ are mutually inverse,

$$T_\mu T_\zeta = T_\zeta T_\mu = \hat{I}, \tag{5.2}$$

this being the classical Möbius inversion formula restated as an operator identity.

5.2. **The commutator.**

Proposition 5.4. *For any arithmetic function f,*

$$[\hat{N}, T_f] = \hat{N}\, T_g, \qquad g(d) = f(d)\left(1 - \frac{1}{d}\right). \tag{5.3}$$

Proof. Directly from Definitions 5.2 and 5.3, for $a \in A$,

$$(\hat{N}T_f a)(n) - (T_f \hat{N} a)(n) = n \sum_{d|n} f(d) a(n/d) - \sum_{d|n} f(d)\, \frac{n}{d}\, a(n/d) \tag{5.4}$$

$$= n \sum_{d|n} f(d) \left(1 - \frac{1}{d}\right) a(n/d) \tag{5.5}$$

$$= n\, (T_g a)(n) = (\hat{N} T_g a)(n), \tag{5.6}$$

with g as in (5.3). □

Corollary 5.5. *Taking $f = \mu$, $g(d) = \mu(d)(d-1)/d$ is not identically zero (e.g. $g(2) = -1/2$), so $[\hat{N}, T_\mu] \neq 0$.*

Remark 5.6 (Consistency check). Since $[\hat{N}, \hat{I}] = 0$, the identity (5.2) and the Leibniz rule for commutators, $[\hat{N}, T_\mu T_\zeta] = [\hat{N}, T_\mu] T_\zeta + T_\mu [\hat{N}, T_\zeta]$, give a relation between the commutators of $\hat{N}$ with T_μ and with T_ζ. This relation has not been developed further here.

5.3. **The escape from Hilbert space.**

Definition 5.7. Let $e_n \in A$ be the indicator sequence, $e_n(k) = 1$ if $k = n$ and 0 otherwise: the position eigenstate at n.

Proposition 5.8.

$$(T_\mu e_n)(m) = \mu(m/n) \cdot [\, n \mid m \,]. \tag{5.7}$$

In particular, $T_\mu e_n$ is supported on the infinite set of multiples of n.

Proof. By (5.1), $(T_\mu e_n)(m) = \sum_{d|m} \mu(d) e_n(m/d)$. The summand is nonzero only when $m/d = n$, i.e. $d = m/n$, which requires $n \mid m$; when this holds, the unique contributing term is $\mu(m/n)$. □

Corollary 5.9. *$T_\mu e_n \notin \ell^p(\mathbb{N})$ for any $1 \le p < \infty$: applying the momentum-type operator T_μ to an exact position eigenstate produces a vector that cannot be normalized in any Hilbert (or, more generally, ℓ^p) space, and exists only in the Fréchet space A of Definition 5.1.*

Proof. By Proposition 5.8, $|(T_\mu e_n)(m)| \in \{0, 1\}$ on the multiples of n, an infinite set on which the entries do not tend to 0; hence the sequence is not in ℓ^p for any finite p. □

Remark 5.10. Corollary 5.9 is the sense in which the non-Hilbert ambient space of Definition 5.1 is not a stylistic choice: it is forced by the requirement that $\hat{N}$ (position, exact ordinal) and T_μ (the Dirichlet-convolution momentum-type operator) act on a common domain. No probability distribution is invoked anywhere in this section; the incompatibility recorded in Corollary 5.9 is a deterministic algebraic fact.

6. A Bost–Connes-type flow on the Fréchet space

This section pushes the construction of Section 5 in the direction of the Bost–Connes quantum statistical mechanical system [8], which is built on a genuine Hilbert space. The goal here is more modest: to see how much of that structure – a Hamiltonian, a time evolution, and a distinguished commutator – survives when the ambient space is the Fréchet space A of Definition 5.1 instead.

6.1. **The ring homomorphism property.**

Proposition 6.1. *For arithmetic functions f, g,*

$$T_f T_g = T_{f*g}, \tag{6.1}$$

where $*$ denotes Dirichlet convolution. In particular, since $*$ is commutative, the operators $\{T_f\}_f$ pairwise commute.

Proof. For $a \in A$,

$$(T_f T_g a)(n) = \sum_{d|n} f(d)\,(T_g a)(n/d) = \sum_{d|n} f(d) \sum_{e|(n/d)} g(e)\, a\Big(\frac{n}{de}\Big) \tag{6.2}$$

$$= \sum_{de|n} f(d) g(e)\, a\Big(\frac{n}{de}\Big) = \sum_{k|n} \Big(\sum_{d|k} f(d) g(k/d) \Big) a(n/k) \tag{6.3}$$

$$= \sum_{k|n} (f * g)(k)\, a(n/k) = (T_{f*g} a)(n). \tag{6.4}$$

□

Remark 6.2. Proposition 6.1 shows that $\{T_f\}_f$ is a commutative subalgebra of operators on A: the non-commutativity found in Proposition 5.4 comes entirely from the position/Hamiltonian-type operator, not from within this family. This mirrors the structure of the Bost–Connes algebra, where the Hecke operators also form a commutative subalgebra and non-commutativity enters only through the number operator [8].

6.2. **Hamiltonian and time evolution.**

Definition 6.3. Define $\hat{H} : A \to A$ by $(\hat{H}a)(n) = \log(n)\, a(n)$ for $n \geq 2$, and $(\hat{H}a)(1) = 0$.

Proposition 6.4. *For $t \in \mathbb{R}$, define formally $\sigma_t(T_f) := e^{it\hat{H}} T_f e^{-it\hat{H}}$. Then*

$$\sigma_t(T_f) = T_{f_t}, \qquad f_t(d) = f(d)\, d^{it}. \tag{6.5}$$

Proof. $(e^{it\hat{H}} T_f e^{-it\hat{H}} a)(n) = n^{it} \sum_{d|n} f(d)\, (e^{-it\hat{H}} a)(n/d) = n^{it} \sum_{d|n} f(d)(n/d)^{-it} a(n/d) = \sum_{d|n} f(d) d^{it} a(n/d) = (T_{f_t} a)(n)$. □

Remark 6.5 (Parallel with Bost–Connes). The twist $f(d) \mapsto f(d)d^{it}$ in (6.5) is the same qualitative operation as the time evolution of the Bost–Connes system, which acts on its generating isometries μ_n by $\sigma_t(\mu_n) = n^{it}\mu_n$ [8]. This is recorded as an adaptation [ADAPT]: same mechanism (multiplicative twist by d^{it}, generated by a logarithmic Hamiltonian), transplanted from a C^*-algebra of isometries on $\ell^2(\mathbb{N})$ to the commutative convolution algebra $\{T_f\}$ on the Fréchet space A. No claim is made that σ_t as defined here is continuous in any topology on A, or that it extends to a genuine one-parameter group of automorphisms of a well-defined algebra; this is left as Question 6.10 below.

6.3. **The commutator with the Hamiltonian.**

Proposition 6.6. *For any arithmetic function f,*

$$[\hat{H}, T_f] = T_{f \log}, \qquad (f \log)(d) := f(d)\log(d). \tag{6.6}$$

Proof. For $a \in A$ and $n \geq 1$,

$$(\hat{H} T_f a)(n) - (T_f \hat{H} a)(n) = \log(n) \sum_{d|n} f(d) a(n/d) - \sum_{d|n} f(d) \log(n/d)\, a(n/d) \tag{6.7}$$

$$= \sum_{d|n} f(d) \big[\log n - \log(n/d)\big] a(n/d) = \sum_{d|n} f(d) \log(d)\, a(n/d) \tag{6.8}$$

$$= (T_{f\log} a)(n). \tag{6.9}$$

□

Corollary 6.7.

$$[\hat{H}, T_\mu] = -\, T_\Lambda\, T_\mu, \tag{6.10}$$

where Λ is the von Mangoldt function.

Proof. The classical identity $\Lambda = \mu * \log$, i.e. $\Lambda(n) = \sum_{d|n} \mu(d) \log(n/d)$, is standard [9, Thm. 2.10]. Expanding $\log(n/d) = \log n - \log d$ and using $\sum_{d|n} \mu(d) = \varepsilon(n)$ (the Dirichlet identity element) gives, for $n \geq 1$,

$$(\mu \log) * \zeta = -\Lambda, \tag{6.11}$$

where $(\mu \log)(d) = \mu(d)\log(d)$. Since $\zeta^{-1} = \mu$ in the Dirichlet ring, (6.11) gives $\mu \log = -\Lambda * \mu$. By Proposition 6.1, $T_{\mu\log} = T_{-\Lambda*\mu} = -T_\Lambda T_\mu$. Combined with Proposition 6.6 at $f = \mu$, this gives (6.10). □

Remark 6.8. Corollary 6.7 refines the bare non-vanishing statement of Corollary 5.5: the commutator of the Hamiltonian $\hat{H}$ with the Möbius operator T_μ is not an arbitrary nonzero operator but is exactly (minus) the von Mangoldt operator composed with T_μ itself. The incompatibility between $\hat{H}$ (energy/ordinal scale) and T_μ (Möbius momentum) is thus governed, in this formalism, by the same function Λ that records the location and multiplicity of prime powers – a fact obtained here as a derived consequence of Definitions 5.2 and 6.3, not assumed.

6.4. **A formal partition function.**

Heuristic 6.9. On $\ell^2(\mathbb{N})$, the Bost–Connes Hamiltonian has partition function $\mathrm{Tr}(e^{-\beta H}) = \sum_n n^{-\beta} = \zeta(\beta)$, a rigorous trace of a well-defined trace-class operator for $\mathrm{Re}\,\beta > 1$ [8]. On the Fréchet space A there is no inner product and no trace-class structure, so no analogous trace exists rigorously here. Formally associating to the diagonal operator $e^{-\beta \hat{H}}$ (with eigenvalues $n^{-\beta}$ on e_n) the generating Dirichlet series $\sum_n n^{-\beta} = \zeta(\beta)$ reproduces the same expression, but this is a bookkeeping device, not a trace of an operator on A in any sense established in this notebook. This is recorded as a heuristic only.

Question 6.10. Is there a topology on (a subspace of) A, coarser than or different from the Fréchet topology of Definition 5.1, in which σ_t of Proposition 6.4 is continuous in t and the operators T_f are continuous, so that Heuristic 6.9 could be replaced by an actual trace or an actual KMS condition?

7. Related structures in the literature

This section records what has and has not been located in the literature, and is not itself a source of new results.

- The classical probabilistic model for μ replaces it by a Rademacher random multiplicative function f, $f(p) = \pm 1$ i.i.d. on primes [6]. A naïve central-limit heuristic for $\sum_{n \le x} f(n)$, of the same flavour as the "fluctuating version" discarded in Remark 2.1, predicts fluctuations of order $\sqrt{x \log\log x}$. This has been shown to be false in the opposite direction: $\mathbb{E}\left|\sum_{n \le x} f(n)\right| \asymp \sqrt{x}/(\log\log x)^{1/4}$ [7], i.e. *better* than square-root cancellation, driven by multiplicative correlations among the $f(n)$ rather than by any added source of uncertainty. This is the opposite of what an add-on uncertainty postulate such as (2.2) would suggest.
- The Hilbert–Pólya program and the Berry–Keating $H = xp$ model [10] pursue a different question – realizing the ordinates of the nontrivial zeros of $\zeta(s)$ as the spectrum of a self-adjoint operator – and are not directly about $\mu(n)$ or about the ordinal–Möbius pairing studied here.
- The construction of Section 5 is structurally closest to the Bost–Connes system [8], whose scaling isometries μ_n and phase-type generators $e(r)$ satisfy a non-commutation relation of the same qualitative type as (5.3) (multiplication/averaging over multiples of n against a phase-type variable). That system is built on a genuine Hilbert space and a C^*-algebra of isometries, motivated by class field theory and KMS states, and does not use the Fréchet-space escape argument of Corollary 5.9.

Remark 7.1 (Novelty check). A literature search was carried out for the specific commutator identity (5.3) and for the escape argument of Corollary 5.9. Neither was located in this exact form. Absence of a search result is weak evidence, not a proof of originality; the pieces used (the Dirichlet convolution ring [5], elementary computation) are all standard.

8. Open questions

Question 8.1. Does the commutator identity (5.3), or the escape argument of Corollary 5.9, connect in any precise way to the Björner cellular complex realization of $M(n)$ as a reduced Euler characteristic, used elsewhere in this research program?

Question 8.2. How does Corollary 4.6 depend on the choice of partition $\Delta_{-1}, \Delta_0, \Delta_{+1}$ in Definition 4.4? Is there a partition-independent formulation of the same complementarity statement?

Conjecture 8.3 (Purely exploratory)**.** *The relation recorded in Remark 5.6 between $[\hat{N}, T_\mu]$ and $[\hat{N}, T_\zeta]$ extends to a full family of commutator identities $[\hat{N}, T_f]$ for f ranging over the multiplicative functions, organized by the Dirichlet-series structure of f. No evidence beyond Proposition 5.4 is offered for this at present.*

Notebook History

Version 0.1: Naive quantum ansatz (MQMNL) recorded and identified as a failed attempt (Section 2).
Version 0.2: Elementary obstruction to exact canonical conjugacy established (Section 3).
Version 0.3: Number-phase Hilbert-space construction added (Section 4).
Version 0.4: Non-Hilbert Dirichlet-convolution construction added, including the escape-from-Hilbert-space argument (Section 5).
Version 0.5: Bost–Connes-type Hamiltonian, time evolution, and commutator with T_μ added, expressing the incompatibility between $\hat{H}$ and T_μ exactly in terms of the von Mangoldt operator; formal partition function flagged as heuristic (Section 6).

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